

lec 6 System Basics

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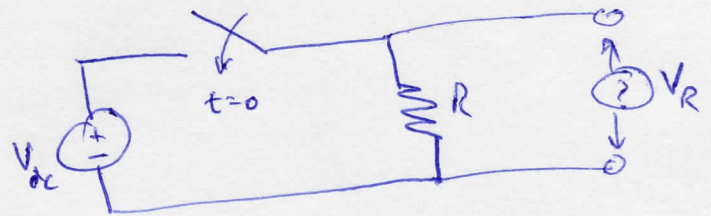
— we have previously looked at signals and several of their types (classifications)

— Signal: collection of data
: carries some information (even noise)

— Today we look at Systems

why? $\leftarrow$ all around us $\leftarrow$ extract or
we often need to modify info in signals

examples $\leftarrow$ car
car
circuit
room



$$V_R = IR$$

— So many Systems!

— How can we know response?

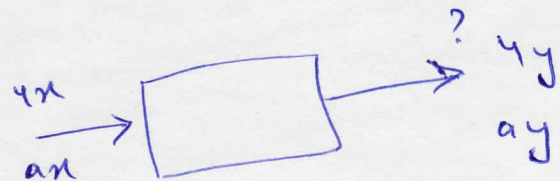
— Some general way of modeling it? (independent of system nature field)

— Perhaps we can study some properties?

→ Probe a block box?

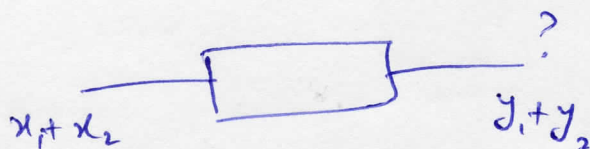
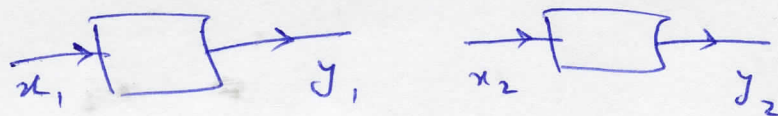


Q. → what happens if I scale the input?



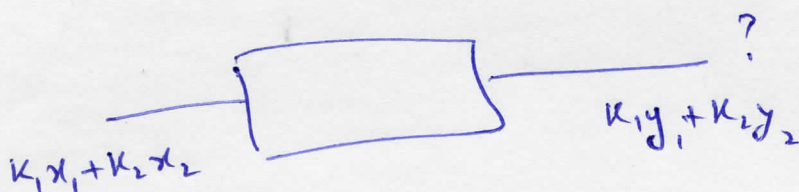
→ if output also scaled by same factor } ← Homogenous

Q. → what happens if I add signals?



→ if output also sum of original outputs } ← Additive

→ we can combine the above two in one form



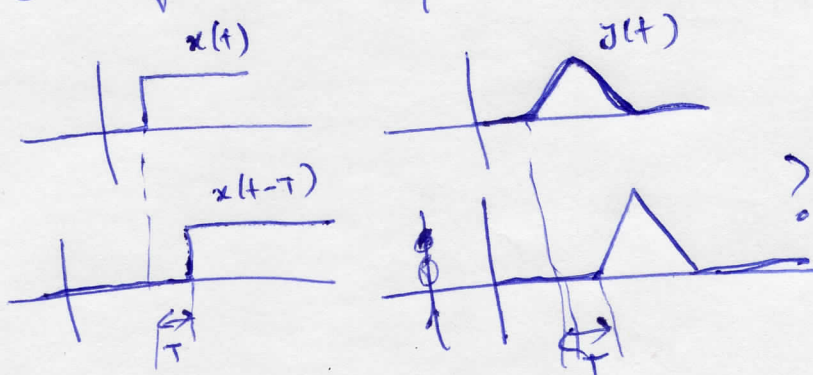
Homogeneous
Additive

→ if yes, then system allows Superposition.

→ And we call such a system "Linear".

→ type fields $\begin{cases} \text{linear} \\ \text{non linear} \end{cases}$

Q. → what happens when input is shifted?

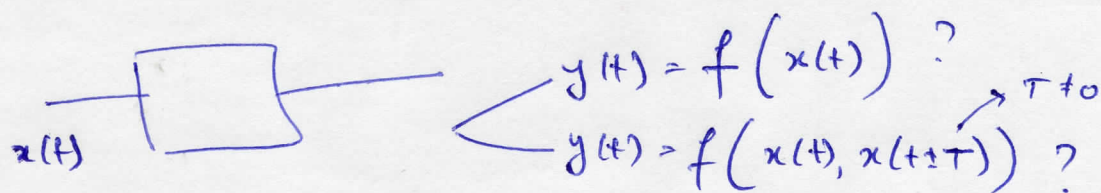


→ if output is also shifted by same amount } ← Time Invariant

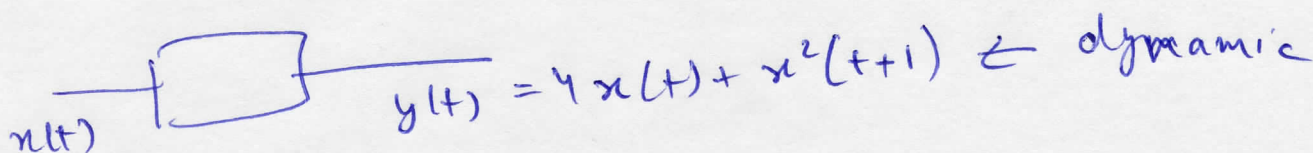
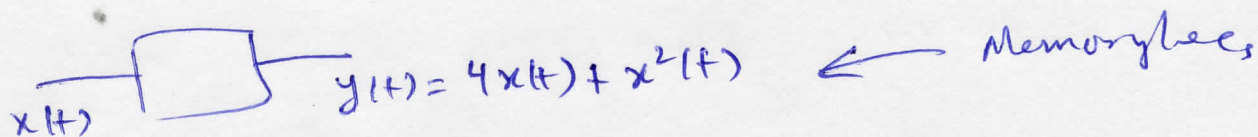
→ Huge classification $\begin{cases} \text{TI} \\ \text{TV} \end{cases}$

→ Huge class: LTI

Q. → Does system output depend on past or future inputs? (or only on current input?)

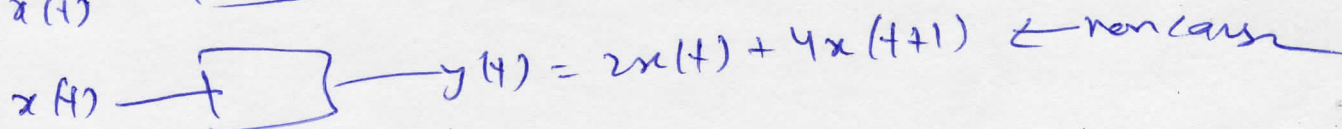
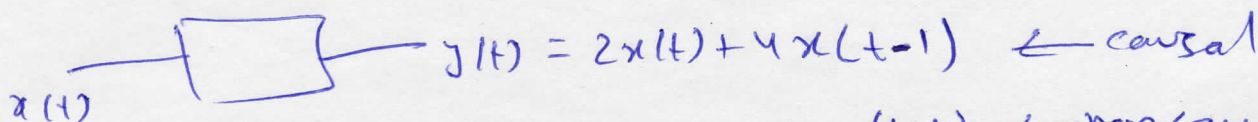


→ if output at time t depends only on $x(t)$ (and not on $x(t \pm T)$ etc.) then → Memoryless (also called Instantaneous)
 → otherwise: Dynamic (or Memory system)



Q. → Does output depend only on current + past inputs?

→ if yes: Causal (all real systems are causal) ← cannot have output before even apply input.



→ why study non-causal?

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→ suppose signal is recorded!

→ then future and past values are available

$$\{x(t_1) \ x(t_2) \ x(t_3) \ x(t_4) \ x(t_5)\}$$

$$y(t_1) = \frac{x(t_1) + x(t_2) + x(t_3) + x(t_4) + x(t_5)}{5}$$

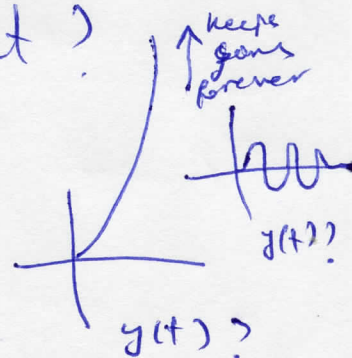
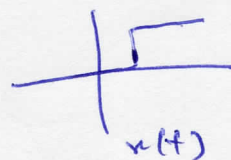
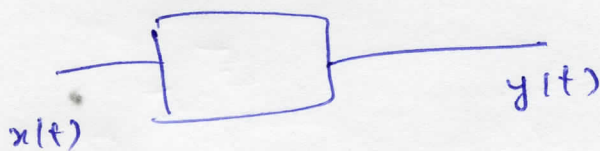
etc.

→ or independent variable can be non-time

$$\begin{array}{cccc} x(l_1) & x(l_2) & x(l_3) & x(l_4) \\ \downarrow & & & \\ y = \frac{x(l_1) + x(l_2)}{2} \end{array}$$

Temperature on wine

Q: Do Bounded inputs give bounded output?

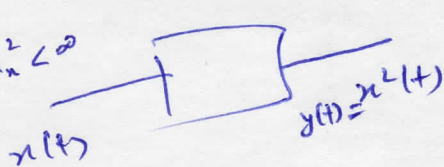


→ if no: system unstable (BIBO-unstable)

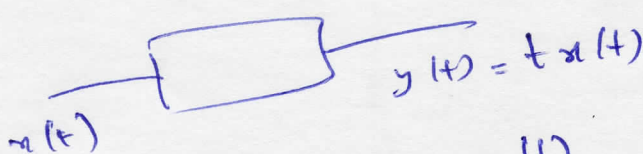
→ if yes: system stable (BIBO-stable)

$$|x(t)| \leq M_x < \infty$$

$$|x^2(t)| = |x(t)|^2 \leq M_x^2 < \infty$$



← BIBO stable



← BIBO unstable

e.g. let $x(t) = u(t)$

Q. → Are inputs outputs continuous or discrete?

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CT System → both input/output CT

DT system → ————— DT

C/D system → input: CT, output DT

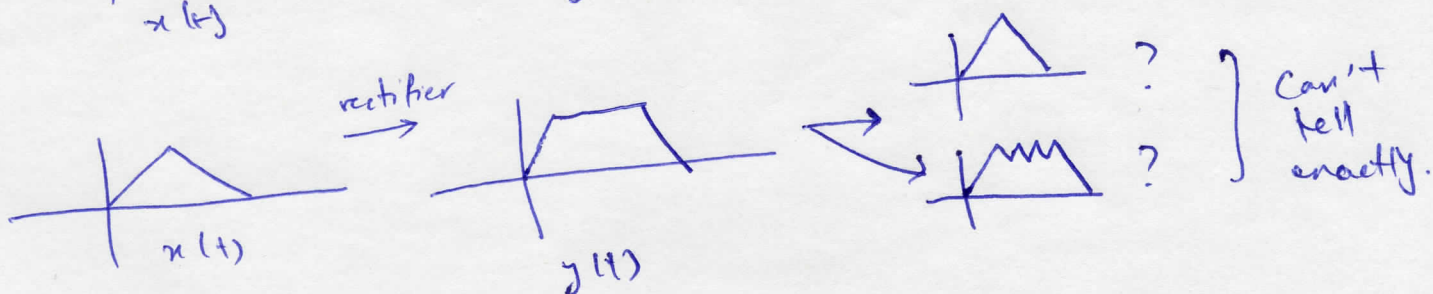
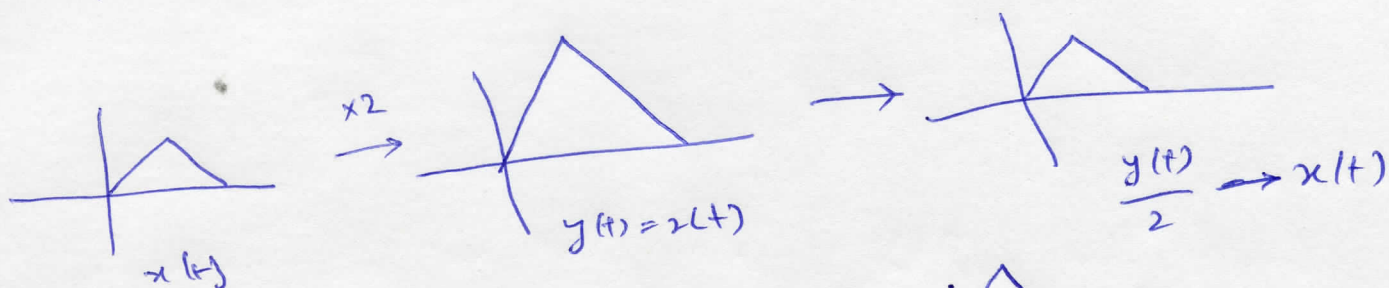
D/C system →

Q. → Are inputs/outputs Analog or digital?

← Analog (both analog)
Digital (both digital)
ADC
DAC

Q. → Can we recover the input exactly from the output? (without knowing input of course)

if yes → Invertible.



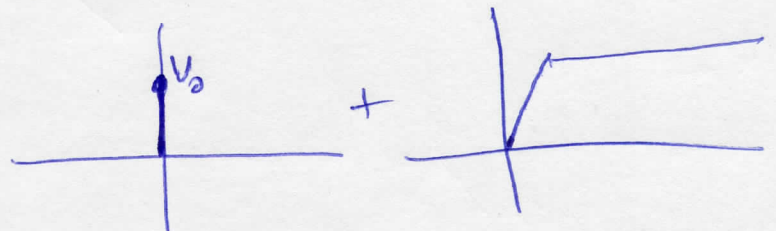
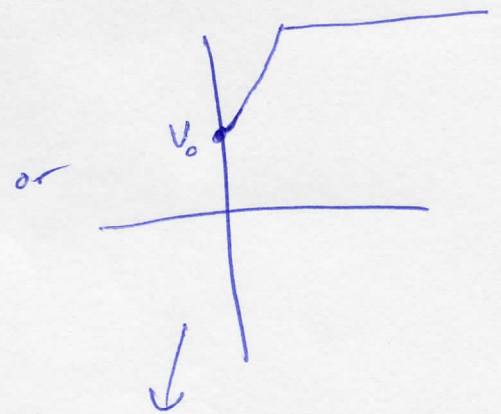
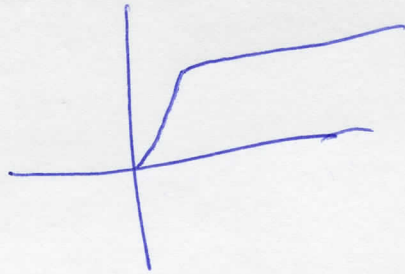
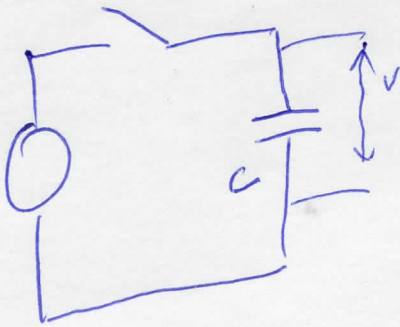
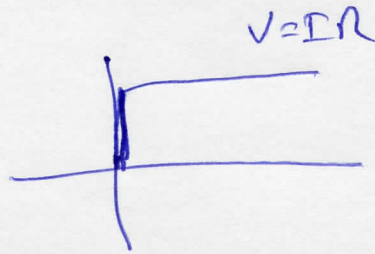
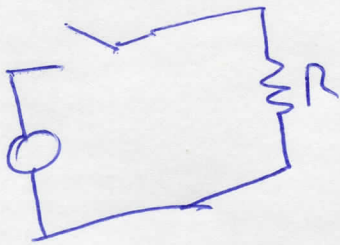
→ SISO, MIMO etc.

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→ Linear System response breakdown

$$\text{Total Resp} = \text{Zero-input resp} + \text{Zero state response}$$

(Zero state = zero initial condition)



Zero input resp + Zero-state response